Multi-Currency Travel Budget Analysis

1. Problem statement

Create a Python program to convert travel expenses in different currencies into Indian rupees using supplied exchange rates. Include conversion fees, compare spending against a budget, and identify the largest expense category.

2. Objectives

- Convert foreign currency expenses into INR.
- Calculate conversion fees.
- Compare total spending with a budget.
- Identify the largest expense category.

3. Assumptions

Sample exchange rates:

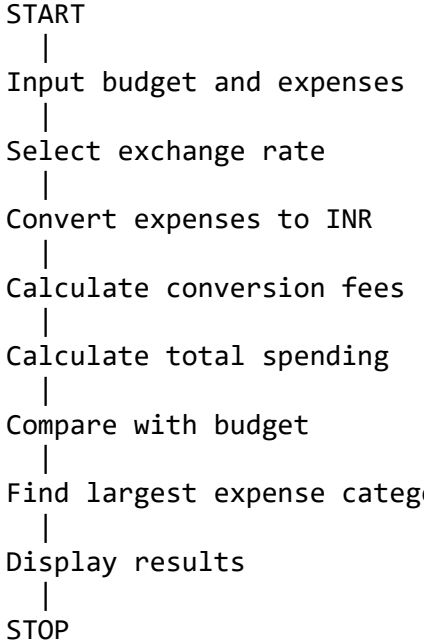
- 1 USD = ₹85.
- 1 EUR = ₹95.
- 1 GBP = ₹110.
- Conversion fee: 2% of converted expenses.
- The travel budget is entered in INR.

These exchange rates are illustrative values for the assignment, not current market rates.

4. Algorithm

1. Input the travel budget.
2. Input expense categories and amounts in foreign currencies.
3. Select the appropriate exchange rate for each expense.
4. Convert each expense to INR.
5. Calculate the conversion fee.
6. Calculate total spending.
7. Compare spending with the budget.
8. Identify the largest expense category.

5. Flowchart



6. Pseudocode

```
START
INPUT budget
rates = USD:85, EUR:95, GBP:110

FOR each expense
    INPUT category, amount, currency
    converted = amount * exchange_rate
    fee = converted * 2 / 100
    total = converted + fee
END FOR

Calculate total spending
IF spending > budget
    PRINT "Over budget"
ELSE
    PRINT "Within budget"
END IF

Find category with largest total
PRINT results
STOP
```

7. Python source code

```
rates = {
    "USD": 85,
    "EUR": 95,
    "GBP": 110
}

budget = float(input("Enter travel budget in INR: "))
n = int(input("Enter number of expense categories: "))

expenses = {}
total_spending = 0

if budget < 0 or n <= 0:
    print("Invalid input")
else:
    valid = True

    for i in range(n):
        category = input("Enter expense category: ")
        currency = input("Enter currency (USD/EUR/GBP): ").upper()
        amount = float(input("Enter expense amount: "))

        if currency not in rates or amount < 0:
            valid = False
```

```
            break

        converted = amount * rates[currency]
        fee = converted * 0.02
        final_amount = converted + fee

        expenses[category] = final_amount
        total_spending += final_amount

    if not valid:
        print("Invalid expense details")
    else:
        largest = max(expenses, key=expenses.get)

        print("Total spending: Rs.", round(total_spending, 2))
        print("Largest expense:", largest)
        print("Largest amount: Rs.", round(expenses[largest], 2))

        if total_spending > budget:
            print("Over budget by Rs.", round(total_spending - budget, 2))
        else:
            print("Remaining budget: Rs.",
                  round(budget - total_spending, 2))
```

8. Test cases

Test	Budget	Expenses	Expected result
1	₹50,000	Hotel: $200	Within budget
2	₹10,000	Hotel: $100, Food: $50	Over budget
3	₹30,000	Travel: €100, Hotel: $200	Identify Hotel as largest
4	₹20,000	Shopping: £50	Within budget